

2.1 The Periodic Table: Chemical Periodicity

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.1 The Periodic Table: Chemical Periodicity
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Sodium sulfide, Na_2S , is a reactive yellow solid, produced when sodium and sulfur react together.

Use of the periodic table is relevant to this question.

How do the ionic radius and atomic radius of sodium compare with those of sulfur?

	Ionic radius	Atomic radius
A	Sodium < Sulfur	Sodium < Sulfur
B	Sodium < Sulfur	Sodium > Sulfur
C	Sodium > Sulfur	Sodium > Sulfur
D	Sodium > Sulfur	Sodium < Sulfur

[1 mark]

Question 2

What happens to the pH of water when magnesium oxide is dissolved into it?

- A. pH increases, because MgO is acidic.
- B. pH decreases, because MgO is basic.
- C. pH increases, because MgO is basic.
- D. Nothing happens because MgO is neutral.

[1 mark]

Question 3

Which observations are made when a sample of phosphorus chloride, PCl_5 , is added to a beaker of water?

- A. No visible changes are observed.
- B. Steamy fumes is the only observation.
- C. A white precipitate is the only observation.
- D. Steamy fumes and a white precipitate are both observed.

[1 mark]

Question 4

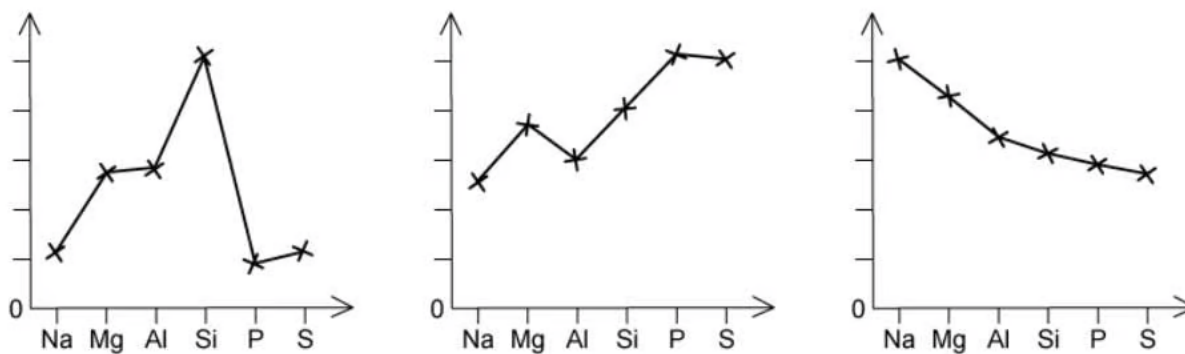
In which species does the underlined atom have an incomplete outer shell?

- A. H_3O^+
- B. CH_3^-
- C. BF_3
- D. MgO

[1 mark]

Question 5

The trends in three physical properties of the elements of period 3 are shown in the graphs.



Which physical property is **not** illustrated?

- A. Electrical conductivity
- B. Atomic radius
- C. Melting point
- D. First ionisation energy

[1 mark]

Question 6

Silicon reacts with chlorine to produce silicon chloride.

How many moles of chlorine are needed to react with 1 mole of silicon?

- A. 2
- B. 3
- C. 4
- D. 5

[1 mark]

Question 7

During the manufacture of steel, the impurity P_4O_{10} is removed by reacting it with calcium oxide. The only product of this reaction is a salt, calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$).

In this reaction, how many moles of P_4O_{10} will react with three moles of calcium oxide?

- A. 0.5
- B. 1.5
- C. 3
- D. 6

[1 mark]

Question 8

Which chlorides of Period 3 elements will form a neutral solution when added to water?

- 1 MgCl_2
- 2 NaCl
- 3 SiCl_4

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Question 9

This question refers to isolated gaseous species.

The species O^{2-} , Ne and Mg^{2+} are isoelectronic.

In which order do their radii increase?

	Smallest → Largest		
A	Mg^{2+}	O^{2-}	Ne
B	Ne	O^{2-}	Mg^{2+}
C	Mg^{2+}	Ne	O^{2-}
D	O^{2-}	Ne	Mg^{2+}

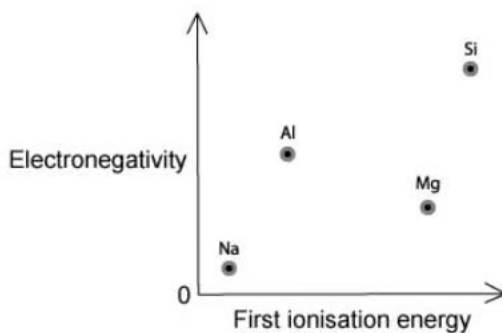
[1 mark]

Question 10

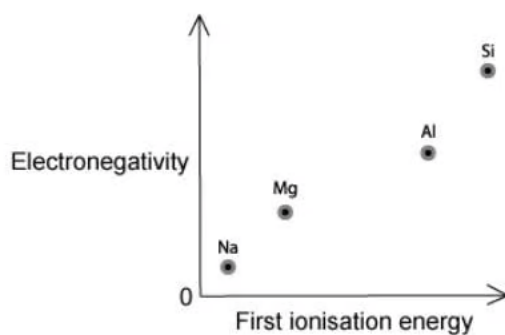
Use of the periodic table is relevant to this question.

Which graph correctly shows the Electronegativity of the elements Na, Mg, Al and Si, from period 3, plotted against their first ionisation energies?

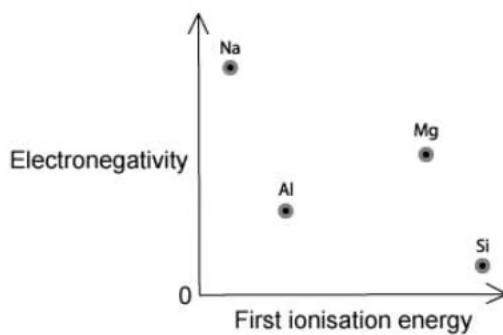
A.



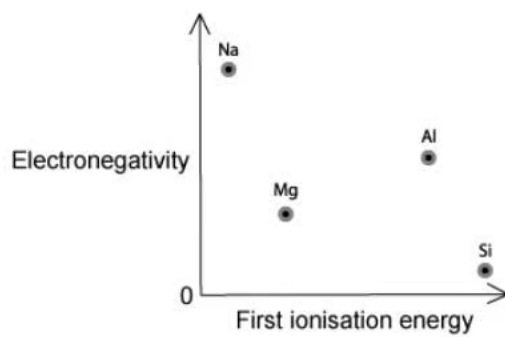
B.



C.



D.



[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Both sodium and sulfur are elements in **Period 3**.
- The **atomic radius** is the distance between the nucleus and the **outermost** electron of an atom.
- The atomic radius is usually measured as $\frac{1}{2}$ **the distance between two nuclei** of two atoms of the same element, in metals, this is the metallic radius and in non-metals the covalent radius.
- The **atomic radius decreases** from left to right across a period ($\rightarrow$).
 - The number of electron shells stays the **same** but the **nuclear charge increases**. Therefore, there is **more attraction** between the nucleus and the outermost electron. Which causes the electrons to be **pulled in** towards the nucleus.
- The **ionic radius increases** from left to right across a period ($\rightarrow$).
 - Groups 1, 2 and 3 form **positive ions**, therefore they **lose the electrons** in their outermost shell and therefore have one less shell of electrons than the atomic form of the element for example,
 - The electronic structure for Na: $1s^2 2s^2 2p^6 3s^1$
 - The electronic structure for Na^+ : $1s^2 2s^2 2p^6$
 - Groups 5, 6 and 7 form **negative ions** as they **gain electrons** in their outermost shell to produce a complete shell and therefore have **one more shell** of electrons than the positive ions of the same period,
 - The electronic structure for S: $1s^2 2s^2 2p^6 3s^2 2p^4$
 - The electronic structure for S^{2-} : $1s^2 2s^2 2p^6 3s^2 2p^6$

2

The correct answer is **C** because:

- The oxides of Period 3 are **basic** on the left-hand side (sodium and magnesium oxides) and **acidic** on the right-hand side (silicon, phosphorus, sulfur and chloride oxides), aluminium oxide is **amphoteric** (can act as an acid and base).
- Magnesium reacts with water to form magnesium hydroxide:
 - Full; $\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2$
 - Ionic; $\text{Mg}^{2+} + \text{O}^{2-} + \text{H}^+ + \text{OH}^- \rightarrow \text{Mg}^{2+} + 2\text{OH}^-$
- **MgO is basic** because it contains an **oxide ion** (O^{2-}) which has a strong tendency to combine with **hydrogen ions** (H^+) to produce hydroxide ions (OH^-).
- The pH of water (~7) increases to 9 when alkaline Mg(OH)_2 is produced.

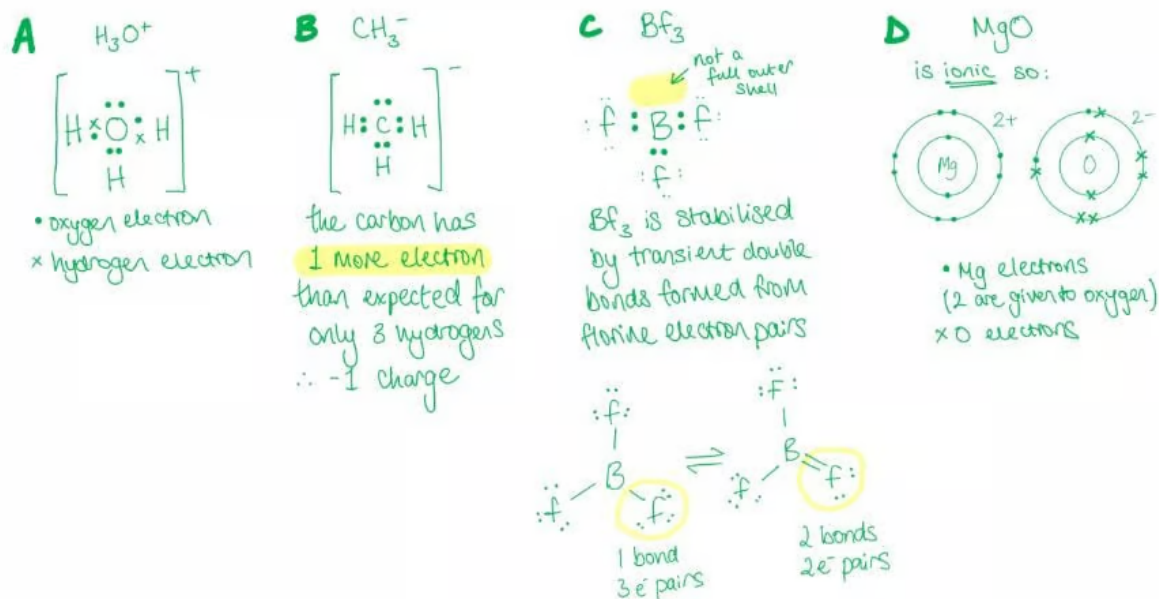
3

The correct answer is **B** because:

- Phosphorus (V) chloride, PCl_5 , is a **covalent molecule**.
- If PCl_5 is added to water there is a violent reaction producing **hydrogen chloride fumes** and an **acidic solution**:
 - $\text{PCl}_5(\text{l}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + \text{HCl}(\text{g})$

4

The correct answer is **C** because:



5

The correct answer is **A** because from left to right the graphs show:

- **Melting point increases across the period from Na to Si then sharply drops.**
 - The melting point of the **metals** increases because the number of electrons **delocalised** from each metal atom increases (1 from Na, 2 from Mg, 3 from Al) so more energy is required to force the delocalised electrons back into the electron shell for solid metals to melt into liquids.
 - Silicon has the highest melting point as it has a **giant covalent structure** with many covalent bonds requiring lots of energy to break.
 - Phosphorus and sulfur can exist as **simple covalent molecules**, with strong covalent bonds within molecules but weak **van der Waals forces** between molecules so much less energy needed to melt solids to liquids.

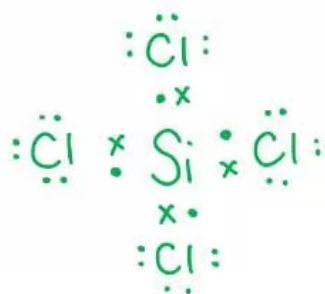
covalent bonds within molecules but weak **van der Waals forces** between molecules so much less energy needed to melt solids to liquids.

- **First ionisation energy** varies, but the general trend is it increases across the period.
 - The ionisation energy depends on the **subshell** the electron is held in.
 - The drop in first ionisation energy from Mg to Al as a result of the electron being removed from the 3p in Al compared to the 3s subshell in Mg (3p is shielded by 3s (less attraction) and so is easier to remove).
- **Atomic radius** decreases from left to right across a period (→).
 - The number of electron shells stays the same, but the **nuclear charge increases**, so **more attraction** between the nucleus and the outermost electron causes the electrons to be pulled in towards the nucleus decreasing the radius.

6

The correct answer is **C** because:

- Silicon has **4 valence** (outer) shell electrons and therefore **requires 4 more** to complete the shell.
- Chlorine has **7 valence** shell electrons and therefore **requires 1 more** to complete the shell.
- One silicon atom **covalently bonds** with 4 chlorine atoms:
 - $\text{Si} + 2\text{Cl}_2 \rightarrow \text{SiCl}_4$



7

The correct answer is **A** because:

- The balanced equation for this reaction is:
 - $\text{P}_4\text{O}_{10} + 6\text{CaO} \rightarrow 2\text{Ca}_3(\text{PO}_4)_2$
- Therefore **1** mole of **P₄O₁₀** reacts with **6** moles of **CaO**.
- So, 3 moles of CaO would react with **0.5 mole of P₄O₁₀**.

8

The correct answer is **B** because:

- MgCl_2 and NaCl/are ionic compounds with **giant ionic structures**.
- Both MgCl_2 and NaCl/dissolves in water to form **neutral solutions**:
 - $\text{MgCl}_2(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{Mg}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq})$
 - $\text{NaCl}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
- $\text{MgCl}_2(\text{aq})$ has a pH between 6.5 – 6.9 and is therefore classed as neutral.
- NaCl/(aq) has a pH of 7 and is therefore neutral.
- They are neutral because the **hydrogen ion concentration**, $[\text{H}^+]$, is **unaffected** as the cation does not alter the $[\text{H}^+]$ and the anion does not attract H^+ from water.

A is incorrect as it does not account for NaCl/also producing a neutral solution.

C & D are incorrect as silicon chloride is a covalent molecule that reacts readily with water to produce hydrochloric acid and silicon oxide: $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$

9

The correct answer is **C** because:

- **Isoelectronic** means that the **number** of electrons and their **configuration** is the **same** in each species: $1\text{s}^2 2\text{s}^2 2\text{p}^6$.
- **Two electrons** have been **added** to the oxygen 2p orbital to produce the **O²⁻ ion**. 2 electrons have been **taken** from the magnesium 3s orbital to produce the **Mg²⁺ ion**.
- As the number of orbitals is the same, the radius is dependent on the **number of protons** in the nucleus. Therefore, the strength of the 'pull' from the nucleus to the

outer electron orbital.

- O^{2-} has 8 protons, Ne has 10 protons, and Mg^{2+} has 12 protons.
- Therefore the **Mg^{2+} nucleus** will have the **greater pull** and create the **smallest nucleus**. Then Ne, O^{2-} will have the smallest nuclear pull and therefore the largest nucleus.

10

The correct answer is **A** because:

- **Electronegativity** is a measure of the tendency of an atom to **attract a bonding pair of electrons**.
- Electronegativity **increases** across a period between group 1 and group 7.
- The number of **electron orbitals** remains the **same** but the **nuclear charge increases** (because the number of protons increases), meaning that there is a **stronger nuclear pull** on the bonding electrons in the outer orbital.
- Therefore, the Electronegativity of the elements in the question is:
 - Na (smallest) < Mg < Al < Si (biggest) hence the y-axis values on the graph
- **First ionisation energy** is the energy required to **remove** one mole of **electrons** (the easiest one to remove) from one mole of **gaseous atoms**:
 - $X(g) \rightarrow X^+(g) + e^-$
- The value of the first ionisation energy across a period depends on whether the electron is **paired or unpaired** within the outer subshell, the **nuclear charge** and **shielding** from other electrons.
- **Sodium** has the lowest ionisation energy because the electron being removed is

pair of electrons.

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- The value of the first ionisation energy across a period depends on whether the electron is **paired or unpaired** within the outer subshell, the **nuclear charge** and **shielding** from other electrons.
- **Sodium** has the lowest ionisation energy because the electron being removed is **unpaired** ($3s^1$) and the nuclear pull is the **lowest** of the 4 elements.
- **Aluminium** has a lower ionisation energy **than magnesium** because there is an increased **distance** from the nucleus
 - Therefore, a **reduction in attraction** to the electron.
 - The 3p orbital is **shielded** from the nuclear pull by electrons in the 3s orbital (as well as others).
 - These factors outweigh the effect of an extra proton.
- **Silicon** has the **highest nuclear charge** and therefore the highest first ionisation energy.